

Code-Layout, Readability and Reusability

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the Human Development Index (HDI) Prediction System based on code organization, readability, and reusability. The project is developed using Python, Flask, HTML, CSS, JavaScript, and Machine Learning. Its modular structure makes the application easy to understand, maintain, and reuse

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation used throughout the code	Yes	Python PEP-8 style followed
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for templates, static, dataset, models, and application
3	Meaningful Variable Names	Variables clearly represent their purpose	Yes	Descriptive variable names used
4	Function / Method Names	Functions are descriptively named	Yes	Functions clearly describe their operations
5	Code Comments	Inline and block comments explain important logic	Partial	Comments added for major sections only
6	Modular Design	Code is divided into reusable modules	Yes	Flask routes, ML model, and utilities are separated

7	No Redundant Code	Duplicate or unused code is removed	Yes	Code optimized by avoiding repetition
8	Error Handling	Exceptions and invalid inputs are handled	Partial	Basic validation and exception handling implemented

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Flask Application (app.py)	Python, Flask	Handles routing and prediction requests	High
2	Linear Regression Model	Scikit-learn, Joblib	Used for HDI prediction	High
3	HTML Templates	HTML, CSS, Jinja2	Home, Prediction, Result, About pages	High
4	Data Preprocessing Module	Python, Pandas	Data cleaning and feature preparation	Medium
5	Prediction Function	Python, Scikitlearn	Generates HDI prediction results	High

Overall Code Quality Assessment

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5	Well-organized project with a clear folder structure
Readability	5	Meaningful naming conventions improve readability
Reusability	5	Modular design enables code reuse across the application
Documentation / Comments	4	Major sections are documented; more comments can improve clarity
Overall Score	4.8/5	The project follows good coding practices and has a maintainable, reusable design suitable for future enhancements.